

ClickHouse

Documentation Infrastructure Audit Report

Prepared by VeriOps

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Risk: Moderate

Key Metrics

Pages scanned: 2302

Report type: Premium Executive Audit

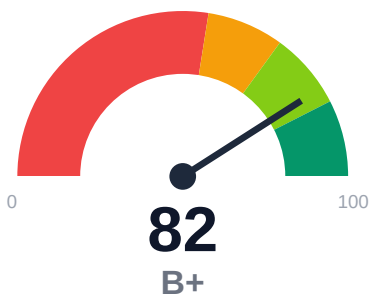
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 508
- Link verification inconclusive for 9 URLs (auth/rate-limit/server block); these are excluded
- SEO/GEO issue rate is high: 17.08%
- API usage-doc coverage is low: 69.7% (1007 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



ClickHouse documentation has measurable quality gaps and structural risks. This audit identifies 4 specific findings across link health, content coverage, and search optimization that directly impact developer experience and support costs.

External posture: SEO/GEO issue rate **17.1%**, public API coverage **69.7%**, broken links **508**.

82 Audit Score	B+ Grade	Moderate Risk Band
2302 Pages Scanned	508 Broken Links	17.1% SEO Issue Rate

Per-Site Breakdown

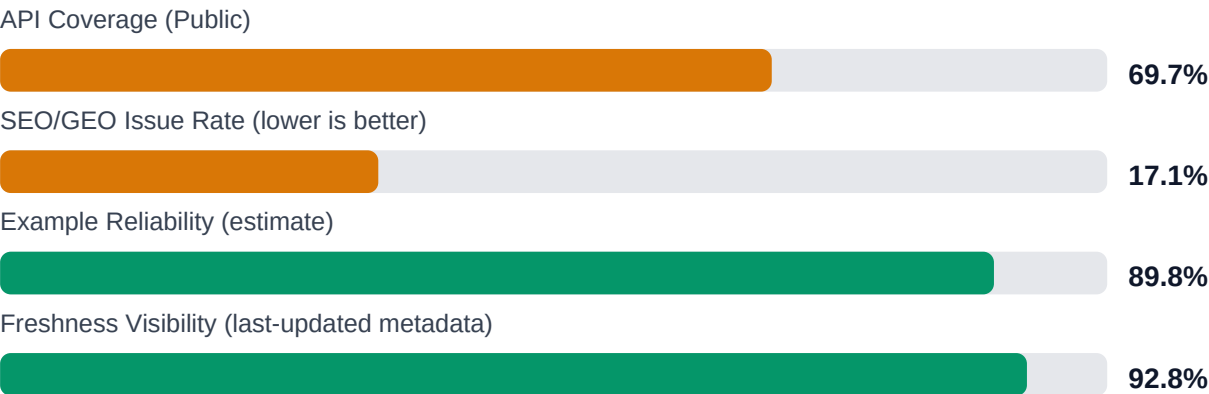
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://clickhouse.com/docs	2302	508	69.7	89.8	17.1

Industry Benchmark Comparison

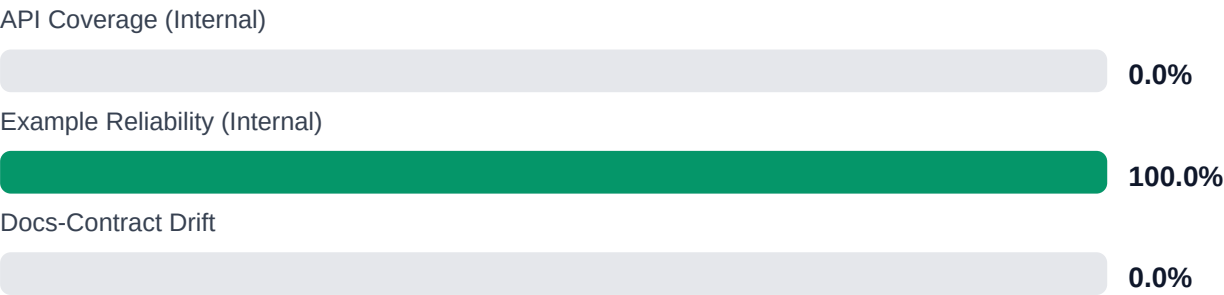
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-10
Industry average (developer docs)	85	-3
Minimum acceptable	70	+12
Your score	82	-

Gap to industry average: **3 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate \$140,822 annual exposure Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	Base Estimate \$249,513 annual exposure Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	High Estimate \$431,728 annual exposure Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo
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Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. 508 broken internal links found. Add pre-commit link checker and redirect map for recently moved pages. Est. remediation: 101 hours.

2. SEO/GEO issue rate 17.1% across 24 automated checks. Top issues: missing meta descriptions, generic headings, low fact density. Run automated SEO optimizer with --fix flag to resolve 80% of issues in one pass.
3. Establish weekly automated audit cadence to track score trajectory. Target: 85+ score within 30 days to match industry average for developer documentation.

Expert Analysis: Strengths and Risks

Link Health

- **508 broken internal links** detected during crawl
- > <https://clickhouse.com/2026/Agenda.aspx>
- > <https://clickhouse.com/2026/DataAILeadershipForum.aspx>
- > <https://clickhouse.com/2026/Program.aspx>
- > <https://clickhouse.com/2026/Register.aspx>
- > <https://clickhouse.com/2026/Showcase.aspx>

SEO/GEO Assessment

- Issue rate: **17.1%** of scanned pages have structural SEO/GEO problems

Content Coverage

- No code examples detected in crawled pages
- Freshness visibility: **93%** of pages expose last-updated metadata

Recommendations

- > Broken internal links: 508
- > Link verification inconclusive for 9 URLs (auth/rate-limit/server block); these are excluded from broken-link counts.
- > SEO/GEO issue rate is high: 17.08%
- > API usage-doc coverage is low: 69.7% (1007 API pages detected)

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.